

Commutative Algebra

Day 3

Exercises (Atiyah - Macdonald)

(7) $\text{Spec}(A)$ is quasi-compact $\rightarrow$

Proof: let $\{U_i\}_{i \in I}$ be an open covering of $\text{Spec}(A)$,
i.e. $\bigcup_{i \in I} U_i = \text{Spec}(A)$. } just use basic elements!
unnecessarily complex.

$$\bigcup_{i \in I} U_i = \text{Spec}(A) \rightarrow \text{each } U_i = \bigcup_{\lambda} X_{i,\lambda}$$

$$\therefore \text{Spec}(A) = \bigcup_{i \in I} \bigcup_{\lambda} X_{i,\lambda}$$

$$\emptyset = \left(\bigcup_{i \in I} \left(\bigcup_{\lambda} X_{i,\lambda} \right) \right)^c = \bigcap_{i \in I} \left(\bigcup_{\lambda} X_{i,\lambda} \right)^c$$

$$= \bigcap_{i \in I} \bigcap_{\lambda} X_{i,\lambda}^c$$

$$= \bigcap_{i \in I} \bigcap_{\lambda} V(i_{i,\lambda})$$

$$= \bigcap_{i \in I} V\left(\bigcup_{\lambda} i_{i,\lambda}\right)$$

$$= \bigcap_{i \in I} V(\langle E_i \rangle) \text{ for some } E_i$$

$$= V\left(\sum_{i \in I} \langle E_i \rangle\right)$$

Now, let $\sum_{i \in I} \langle E_i \rangle = J$, an ideal.

Claim: $1 \in J$

Proof: suppose not. Then, by Krull's $\exists$ maximal ideal $\mathfrak{m} \supseteq J$, but as $\mathfrak{m}$ is prime, hence $\mathfrak{m} \in V(J) \Rightarrow \in$

Thus, $1 \in J$.

As $1 \in J$, this means that there are finitely many $x_1, x_2, \dots, x_n \in J$ $\exists \sum_{1 \leq i \leq n} x_i a_i = 1$, $a_i \in A$.

But then, $V((x_1) + (x_2) + \dots + (x_n)) = V((1)) = \emptyset$

$$\text{Thus, } \emptyset = V(\sum (x_i)) = \bigcap V((x_i)) \\ = \bigcap V(x_i)$$

$$= \bigcap X_{x_i}^c$$

$$= \left(\bigcup X_{x_i} \right)^c$$

$$\Rightarrow \bigcup_{1 \leq i \leq n} X_{x_i} = \text{Spec } A, \text{ as } X_{x_i} \text{ are basic}$$

elements, & $X_{x_i} \subset U_{k_i}$ for some k_i ($1 \leq i \leq n$),

hence this is the required finite subcover.

Thus, $\text{Spec}(A)$ is quasi-compact.

(vi) Each X_f is quasi-compact.

Proof: As $B = \{x_i : a_i \in A\}$ is a basis that generates $\text{Spec}(A)$,
hence, $X_f = \bigcup_{\lambda \in J} X_{f_\lambda}$. If for any $\lambda' \in J$, $f_{\lambda'} = f$,

we are done. Hence, suppose that $f_\lambda \neq f \forall \lambda \in J$.

$$\text{Thus, } X_f^c = \bigcap_{\lambda \in J} X_{f_\lambda}^c \Leftrightarrow V((f)) = \bigcap_{\lambda \in J} V((f_\lambda)) \\ = V\left(\sum_{\lambda \in J} (f_\lambda)\right)$$

$$\text{Let } \sum_{\lambda \in J} (f_\lambda) = I$$

$$\text{Then, } V((f)) = V(I) \Rightarrow \sqrt{(f)} = \sqrt{I}$$

then, $f \in (f) \in \sqrt{(f)} = \sqrt{I}$, i.e. $\exists n > 0 \ni f^n \in I$.

$$\text{i.e. } f^n \in \sum_{\lambda \in J} (f_\lambda)$$

This means there are finitely many indices $\ni$

$$f^n = \sum_{1 \leq i \leq k} f_{\lambda_i} a_i$$

$$\text{Hence, } f^n \in (f_{\lambda_1}) + (f_{\lambda_2}) + \dots + (f_{\lambda_k})$$

If $\mathfrak{p}$ is a prime ideal $\ni \mathfrak{p} \ni f^n \Leftrightarrow \mathfrak{p} \ni f$

$$\text{then } \mathfrak{p} \supseteq (f_{\lambda_1}) + (f_{\lambda_2}) + \dots + (f_{\lambda_k}),$$

$$\text{And if } \mathfrak{p} \supseteq (f_{\lambda_1}) + \dots + (f_{\lambda_k}) \Rightarrow \mathfrak{p} \ni f^n \Leftrightarrow \mathfrak{p} \ni f$$

$$\text{Thus, } V(f) = V\left(\sum_{1 \leq i \leq k} (f_{\lambda_i})\right) = \bigcap_{1 \leq i \leq k} V(f_{\lambda_i})$$

$$\text{Thus, } X_f = \bigcup_{1 \leq i \leq k} V(f_{\lambda_i})^c = \bigcup_{1 \leq i \leq k} X_{f_{\lambda_i}}$$

Hence, we have obtained a finite sub cover.

18) (Notation $\rightarrow$ denote ' x ' as the point in $\text{Spec}(A)$ by $\mathfrak{p}_x$)

(i) $\{x\}$ is closed in $\text{Spec}(A) \Leftrightarrow \mathfrak{p}_x$ is maximal.

Proof:-

($\Rightarrow$) say $\{x\}$ is closed in $\text{Spec}(A)$. Thus, $\{x\} = V(\mathfrak{a})$ for some ideal $\mathfrak{a}$. i.e., the only prime ideal containing $\mathfrak{a}$ is $\mathfrak{p}_x$. As $\mathfrak{a} \neq (1)$ [as then $V(\mathfrak{a}) = \emptyset$], hence by Krull, $\mathfrak{a} \subseteq \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$ in A , however, as $\mathfrak{m}$ is also prime, $\mathfrak{m} = \mathfrak{p}_x$, hence $\mathfrak{p}_x$ is maximal.

($\Leftarrow$) suppose $\mathfrak{p}_x$ is maximal.

Then, as $\mathfrak{p}_x$ is maximal, hence if any other prime $\mathfrak{p}$ exists $\ni \mathfrak{p}_x \supsetneq \mathfrak{p}$, then $\mathfrak{p} = (1)$, which is false.

Hence, $\{\mathfrak{p}_x\} = V(\mathfrak{p}_x)$, hence $\{x\}$ is closed.

(ii) Show $\overline{\{p_n\}} = V(p_n)$

$$\overline{\{p_n\}} = \bigcap_{V(E) \ni p_n} V(E) \quad (\text{intersection of all closed sets containing } p_n)$$

Claim: $\bigcap_{V(E) \ni p_n} V(E) = V(p_n)$

Proof: Let $p \in \bigcap_{V(E) \ni p_n} V(E)$. Thus, $p \in V(E) \forall p_n \in V(E)$.

Can be heavily reduced, as:

trivially

$$\{p_n\} \subseteq V(p_n)$$

as $V(p_n)$

is itself a closed set &

$$p_n \in V(p_n)$$

$$\text{i.e., } p \supseteq E \forall p_n \supseteq E$$

$$\text{i.e., } V(p) \subseteq V(E) \forall V(p_n) \subseteq V(E)$$

$$\text{i.e., } \bigcup_{p_n \in V(E)} V(p) \subseteq V(E)$$

sub claim: $p \in V(p_n)$.

Proof: Suppose not, then, $p_n \not\subseteq p$.

setting $E = p_n$, we have $E = p_n \subseteq p$, but $p_n \not\subseteq p \Rightarrow$

thus, $p \in V(p_n)$

hence, $\bigcap_{V(E) \ni p_n} V(E) \subseteq V(p_n)$

For the other inclusion, let $p \in V(p_n)$.

Claim: $p \in \bigcap_{E \subseteq p_n} V(E)$.

Proof: it's $p \in V(p_n) \Rightarrow p_n \subseteq p$.

Let $E \subseteq p_n$. Then, as $p_n \subseteq p \Rightarrow E \subseteq p \Rightarrow p \in V(E)$,

thus, as $E \subseteq p_n$ was arbitrary, we have $p \in \bigcap_{E \subseteq p_n} V(E)$,

thus, we have $V(p_n) \subseteq \bigcap_{E \subseteq p_n} V(E)$.

Combining, we have $\overline{\{p_n\}} = \bigcap_{p_n \in V(E)} V(E) = V(p_n)$

18) (ii) $y \in \overline{\{x\}} \Leftrightarrow p_x \subseteq p_y$

Proof: As $\overline{\{x\}} = V(p_x)$, hence, $p_y \in V(p_x) \Rightarrow p_x \subseteq p_y$.
Conversely, if $p_x \subseteq p_y \Rightarrow p_y \in V(p_x) = \overline{\{x\}}$.

(iv) $\text{Spec}(A)$ is a T_0 space

Proof: Let $p_x \neq p_y$ be points in $\text{Spec}(A)$.

T.S.: Either there is open $U_x \ni p_x$ s.t. $p_y \notin U_x$ or
(
open $U_y \ni p_y$ s.t. $p_x \notin U_y$.
)
Suppose every open $U_x \ni p_x$ contains p_y & $U_y \ni p_y$ contains p_x .

Thus, every open set containing p_x contains p_y & vice versa
(p_x & p_y are topologically indistinguishable!)

This means every closed set also contains p_x & p_y simultaneously.

As $\overline{\{p_x\}}$ is closed, hence $p_y \in \overline{\{p_x\}}$ & $p_x \in \overline{\{p_y\}}$,
thus, by (iii), $p_x \subseteq p_y$ & $p_y \subseteq p_x \Rightarrow p_x = p_y$,
a contradiction as we assumed $p_x \neq p_y$.

Thus, $\text{Spec}(A)$ is T_0 .

Much better proof ↪

1. Let $p_x \neq p_y$. Then, as ideals, one is not contained in the other.
2. Without loss of generality, assume $p_x \not\subseteq p_y$.
3. By your previous result (part iii), this means $p_y \notin \overline{\{p_x\}}$.
4. Therefore, the set $U = \text{Spec}(A) \setminus \overline{\{p_x\}}$ is an open set.
5. $p_y \in U$ (since it's not in the closure) and $p_x \notin U$ (since it is in the closure).
6. Thus, the space is T_0 .

Q) But is $\text{Spec}(A)$ T_1 as well?

Answer: No, & almost never! (Take prime ideals strictly contained in the other). (For it to be T_1 , we need single points to be closed sets $\rightarrow$ & that only happens if there is only one prime ideal in the Ring).

19) $\text{Spec}(A)$ is irreducible $\Leftrightarrow \text{Nil}(A)$ is prime.

Proof: $(\Rightarrow)$ $X_f \cap X_g = X_{fg} \neq \emptyset \quad \forall f, g \in A$.

$$\text{Nil}(A) = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p} \quad , \quad \text{let } xy \in \text{Nil}(A) = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$$

This means $xy \in \mathfrak{p}$ for all prime $\mathfrak{p}$.

Thus, $V(xy) = \text{Spec}(A)$, & hence $X_{xy} = \emptyset$.

Suppose $\exists$ prime $\mathfrak{p}_0$ s.t. $x \notin \mathfrak{p}_0$ & $y \in \mathfrak{p}_0$.

Hence, $\mathfrak{p}_0 \notin V(x)$, $\mathfrak{p}_0 \in V(y) \Rightarrow \mathfrak{p}_0 \in X_x \cap X_y = X_{xy}$

This means X_{xy} is non-empty, & so is $X_x \& X_y$.

However, $X_x \cap X_y$ is empty, a contradiction to our assumption.

Hence, $\text{Nil}(A)$ must be prime.

$(\Leftarrow)$ Suppose $\text{Nil}(A)$ is prime.

As $\{X_f : f \in A\}$ is a basis which generates $\text{Spec}(A)$, hence enough to show that $X_f \cap X_g = X_{fg}$ is non-empty if X_f & X_g are non-empty.

As $X_{fg} = V(fg)^c$, suppose FTSOC that X_f & X_g are non-empty but $X_{fg} = \emptyset$,
 $V(fg) = \text{Spec}(A)$

Hence, all prime ideals in A contain fg ,

$$fg \in \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p} = \text{Nil}(A),$$

As $\text{Nil}(A)$ is prime, either $f \in \text{Nil}(A)$ or $g \in \text{Nil}(A)$.
 If $f \in \text{Nil}(A)$, this means $V(f) = \text{Spec}(A) \Leftrightarrow X_f = \emptyset$,
 a contradiction since we assumed X_f is non-empty.
 similar for X_g .

Thus, $X_f \cap X_g = X_{fg}$ is always non-empty
 whenever X_f, X_g are, hence $\text{Spec}(A)$ is irreducible.

Modules

See definitions

- (i) A ring, then for modules M & N , $\text{Hom}_A(M, N)$ is
 the module containing all homomorphisms $f: M \rightarrow N$.
 $\ni (f+g)(m) = f(m) + g(m),$
 $f(am) = a \cdot f(m), a \in A.$

- (i) Let $u: M' \rightarrow M$ & $v: N \rightarrow N''$
 $u \in \text{Hom}(M', M)$ & $v \in \text{Hom}(N, N'')$

Then, this leads to $\bar{u}: \text{Hom}(M, N) \rightarrow \text{Hom}(M', N)$
 & $\bar{v}: \text{Hom}(M, N) \rightarrow \text{Hom}(M, N'')$

defined by $\bar{u}(f) = f \circ u$, $\bar{v}(f) = v \circ f$
 $\hookrightarrow$ these are A -module homomorphisms.

Summary

(i) Further exploration into Zariski Topology

- $\text{spec}(A)$ is quasi-compact
- $\forall f \in A, X_f \in \mathcal{B}$ is quasi-compact
(Write $U = \bigcup_{f \in \lambda} X_f$, use radicals & $V(E)$ reverse inclusions $\rightarrow$ conclude)
- If A is noetherian, then above properties hold as every ideal is finitely generated. 

(ii) Next, we treat $\mathfrak{p}_x \in \text{spec}(A)$ as points, & exploit them $\rightarrow$

(Notationally, treat $x \in \text{spec}(A) \Leftrightarrow \mathfrak{p}_x$) $\mapsto$

- $\{x\}$ is closed iff $\mathfrak{p}_x$ is a maximal ideal.
- $\overline{\{x\}} = V(\mathfrak{p}_x)$.
- $y \in \overline{\{x\}} \Leftrightarrow \mathfrak{p}_x \subseteq \mathfrak{p}_y$.
- Conclusion that $\text{spec}(A)$ is T_0 space,
& it is almost NEVER, as it only happens if A has only one prime ideal. (& A is local)
- $\text{spec}(A)$ is irreducible iff $\text{Nil}(A)$ is a prime ideal.

(iii) Basic Intro to Modules, $\text{Hom}_A(M, N)$ & derived modules from $u: M' \rightarrow N$ & $v: M \rightarrow N''$.
